

Q-Orbit — Manuscript Claim Ledger

Document ID: QO-LEDGER-001 | **Version:** 1.0 | **Date:** 2026-08-27 **Applies to:** Q-Orbit manuscript V1.0-RC2 (Deliverable D7). The manuscript's §20 appendix is a condensed rendering of this ledger; this document is authoritative. **Boundary:** THEORETICAL / NOT PHYSICALLY VALIDATED. Every claim below is a theoretical/computational claim about a frozen software fixture and its documentation. No claim in this ledger is physical, device-security, mission, procurement, or deployment evidence. Numerical reproduction is not validation. **Source policy:** numbers are drawn only from the Canonical Facts Record (CFR v1.1, QO-CFR-001), the Phase 1 Canonical Audit (D1/D2), the Device-Imperfection Mapping (D5), the Proof-Profile Comparison (D4), the Literature and Proof Review (D3), the V0.17-TA1 Architecture Specification (D6), the Phase 1 Closure Record, and the controlled artifacts at `/mnt/agents/output/extracted/v016/` (01–10). Nothing outside this package was used.

1. Controlled claim labels (definitions)

Label	Meaning
NUMERICALLY-VERIFIED	Recomputed bit-exactly (or to stated tolerance) from hash-verified controlled artifacts; deterministic model output, not physical measurement.
THEORETICALLY-SUPPORTED	Follows from a verified mathematical structure / published proof (e.g. Lim et al. 2014, Sidhu et al. 2022); transcribed faithfully into the fixture; validity conditional on the proof's own assumptions.
LITERATURE-SUPPORTED	Supported by the verified citation record (D3); context only, never device evidence.
ASSUMPTION-DEPENDENT	Conditional on stated model assumptions (perfect phase randomization, IID pulses, exact intensities, squashing, no side channels, point parameters) that are currently unmeasured.
CHARACTERIZATION-REQUIRED	Cannot be instantiated numerically until physical characterization exists; symbolic only.
PHYSICAL-VALIDATION-REQUIRED	Would require an experimental/physical validation activity that has not been performed.
OPERATIONALLY-UNSUPPORTED	No artifact supports an operational reading; any operational interpretation is out of scope of the evidence.
BLOCKED	Claim class is prohibited/fail-closed; unattainable in current scope (EV-10); gated by an open REQ item or by the disclosed-limitations register.

Evidence classes (CFR v1.1): EV-1a HASH-VERIFIED-ARTIFACT ·EV-1b CSV-RECOMPUTED ·EV-1c HASH-CHAIN-LISTED ·EV-2 INVARIANT ·EV-3 CROSS-ARTIFACT ·EV-4 SINGLE-ARTIFACT ·EV-5 LITERATURE-SUPPORTED ·EV-9 UNVERIFIED-PENDING-CONTROLLED-INPUTS ·EV-10 BLOCKED.

2. Claim ledger

Column key — Label: NV = NUMERICALLY-VERIFIED, TS = THEORETICALLY-SUPPORTED, LS = LITERATURE-SUPPORTED, AD = ASSUMPTION-DEPENDENT, CR = CHARACTERIZATION-REQUIRED, PVR = PHYSICAL-VALIDATION-REQUIRED, OU = OPERATIONALLY-UNSUPPORTED, BL = BLOCKED. Manuscript sections refer to the actual V1.0-RC2 section numbering (QA-corrected 2026-08-27 — verifier finding 5.1: an earlier revision of this ledger carried the superseded rendering-B numbering; all pointers re-mapped and verified against the RC2 section table).

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-01	Baseline optimal half-window is 102 s (205 sample bins; edge elevation 30.4813547009598°)	§11.1	NV	CFR N-01; EV-1a (artifact 03) + EV-1b (artifact 02)	Deterministic model output on the frozen 693-sample loss curve.
CL-02	Baseline signed key margin M = 41,338.62418456675 bits	§9.3, §11.1	NV	CFR N-02; EV-1a; margin identity recomputed bit-exact (diff 0.0)	Upper bound w.r.t. error-correction efficiency (see CL-08). Conditional computation; no security claim.
CL-03	Floored candidate key = 41,338 bits = floor(max(M,0))	§11.1	NV	CFR N-03; EV-1a + EV-2	Floor convention verified on baseline and on all 1,681 grid rows. Not a released key (see CL-56).
CL-04	X-basis QBER = 0.017422686665352745 (= m_X/n_X, bit-exact)	§11.1	NV	CFR N-04; EV-1a + recomputation	Model QBER, not measured.
CL-05	Phase-error bound _X = 0.09270161340569935	§9.3, §11.1	NV	CFR N-05; EV-1a	0.5 cap not hit at baseline; binding in 1,088/1,681 grid rows (EV-1b).
CL-06	Single-photon yield estimate s_X,1 = 183,803.04893680647	§11.1	NV	CFR N-06; EV-1a + EV-3 (REG-007, AUD-016)	Decoy-estimator output of the frozen fixture.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-07	Baseline count observables: $n_X = 492,818.0901525894$; $n_Z = 48,555.17200782972$; $m_X = 8,586.215167746126$; $s_X,0 = 5,047.784882329125$; $s_Z,1 = 12,007.470453438744$; $v_Z,1 = 845.9615478747239$	§11.1	NV	CFR N-16...N-20; EV-1a (artifact 03)	$s_X,0$ single-artifact within the run summary (EV-1a record); all feed the bit-exact margin identity.
CL-08	$_EC = 65,385.40180119235$ bits (binomial-ppf logM construction)	§9.3	NV	CFR N-21/N-27; EV-1a + EV-2	Corresponds to ideal $f_EC = 1$ leakage accounting; realistic $f_EC > 1$ all margins/keys are UPPER BOUNDS w.r.t. EC efficiency. $f_EC \sim 1.16$ is literature context only (EV-5), never a Q-Orbit value.
CL-09	Finite penalty $6 \cdot \log_2(21/_s) + \log_2(2/_c) = 256.5669430839006$ bits	§9.3	NV	CFR N-22; EV-1a	Pair ($_s=1e-10$, $_c=1e-9$) reproduces it bit-exactly but is one of infinitely many solutions; individual epsilons EV-9 (gated by REQ-01).
CL-10	Probability-weighted mean photon number = 0.62400964	§11.1	NV	CFR N-23; EV-1a	Constituent p, live in the missing V0.6 manifest (EV-9; REQ-01).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-11	Grid partition: 568 positive / 1,113 nonpositive / 1,681 total	§11.3	NV	CFR N-07; EV-1b + EV-1a + EV-2	UPPER- ENVELOPE quantity: half-window re-optimized per point (CL-22); the positive count at any fixed window is 568. Deterministic screen partition, not a probability.
CL-12	Positive grid fraction = 0.33789411064842356 (= 568/1,681 exactly in binary64)	§11.3	NV	CFR N-08; EV-1b + EV-1a + EV-2	Same upper-envelope disclosure as CL-11. Not the fraction of any fixed-window configuration; never a probability, reliability, availability, or yield.
CL-13	Grid median signed margin = −2,624.946810258186 bits	§11.3 (and abstract)	NV	CFR N-09 + C-01; EV-1b + EV-1a + EV-2	NEGATIVE. Sign forced by the partition invariant (1,113 nonpositive of 1,681). Rendering A' s positive table value was a sign flip, corrected (C-01). Never render positive.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-14	Grid minimum signed margin = $-3,828.414517626367$ bits (cell SCR-1681 at $(2e-6, 0.015)$, half-window 1, capped at 0.5)	§11.3	NV	CFR N-10 + C-02; EV-1b + EV-1a + EV-2	NEGATIVE. Rendering A' s positive minimum was internally inconsistent with the 568/1,113 split; corrected (C-02). Never render positive.
CL-15	Grid maximum signed margin = $+142,540.7481180454$ bits (cell SCR-0001 at $(1e-7, 0.003)$, half-window 221)	§11.3	NV	CFR N-11; EV-1b + EV-1a	The only positive extremal statistic; never disputed.
CL-16	Eight normalized local responses (rank order): additional loss -0.08215788925285143 ; detector efficiency $+0.035673639848324994$; repetition rate $+0.01946358671353013$; extraneous count -0.016195046443390957 ; intrinsic QBER -0.006768764678164013 ; signal intensity -0.005861881586843529 ; weak decoy -0.0007369594599526124 ; afterpulse -0.0006325501181450469	§11.2	NV	CFR N-12; EV-1b (arithmetic re-verified $1e-12$) + EV-1c	Artifact 06 claim_class: LOCAL-NUMERICAL-RESPONSE-NOT-PHYSICAL-SENSITIVITY. Modeled responses of the fixture, not measured device sensitivities. Windows re-optimize at perturbation points (e.g. loss ± 0.1 dB $\rightarrow$ 104/101 s).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-17	Zero-key frontier set: 16 rows (8 parameters $\times$ 2 sides), 10 CROSSING-FOUND, 6 NO-CROSSING-IN-DECLARED-DOMAIN; e.g. loss HIGH 14.507927510764345 dB; extraneous HIGH 8.958206093312436e-07; intrinsic-QBER HIGH 0.013335017073411072	§11.4	NV	CFR N-24; EV-1c + EV-3 (07 audit 08 AUD-016-005/006)	Software-fixture frontiers only; each row carries “not a hardware acceptance threshold”. Not device tolerance data.
CL-18	Regression suite: 12 tests, 12 PASS (REG-001...012)	§10.6	NV	CFR N-13; EV-1c (04) + EV-1a (03) + EV-3 (08)	Includes +20 dB negative test, grid identity, grid partition, 2 boundary invariants. Computational verification, not physical.
CL-19	Independent package audit: 20 checks, 20 PASS (AUD-016-001...020)	§10.6	NV	CFR N-14; EV-1c (08)	All 20 payloads cross-consistent with artifacts 03/04/05/06/07.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-20	V0.13 independent comparison: 50 locked vectors / 971 metrics, zero open numerical discrepancies	§10.6	NV	CFR N-25; EV-3 (V0.13 record manuscript §3.6); underlying JSON EV-9	The V0.13 JSON itself is not in the bundle (EV-9); claim rests on cross-artifact wording, not re-verification. All 12 V0.13 limitations remain open.
CL-21	Window rule: integer half-window sweep with argmax objective, smaller window breaks ties; observed extremes max 221 s (cross-artifact), min 1 s (artifact 05 only; 07' s minimum is 67 s)	§10.2	NV (rule/ extremes) / AD (1–221 s bound)	CFR B-04; EV-3 (rule, max window) + EV-4 (min window) + EV-9 (config bound)	The 1–221 s sweep bound is a controlled-input value not in the bundle (REQ-01); stated by corroboration only.
CL-22	Half-window is re-optimized per evaluated point in sensitivity, screen, and frontier computations	§10.2, §11.2–4.3	NV	CFR N-26; EV-1b (06 window columns; 05 optimized_half_window_s)	Basis of the upper-envelope disclosures in CL-11/CL-12.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-23	Screen construction: 40-point linspace over each V0.7 range baseline $\rightarrow$ 41 unique values per axis $\rightarrow$ 1,681 points; ranges are engineering bounds, not distributions	§11.3	NV	CFR B-05; EV-1c (01) + EV-1b (05); axis reconstruction exact	Axes: extraneous $1e-7 \cdots 2e-6$ $\{5e-7\}$; intrinsic QBER $0.003 \cdots 0.015$ $\{0.005\}$. Range provenance vs the V0.7 register is EV-9 (REQ-03).
CL-24	Key convention: per-pass signed margin retained; candidate key = $\text{floor}(\max(\text{margin}, 0))$	§9.3	NV	CFR B-06; EV-1c + EV-1a	Signed margins are never reported as keys without the floor/max step.
CL-25	Baseline channel/ model parameters: additional_system_loss_db = 13.0 dB; rep rate $1e8$ Hz; $p_{ec} = 5e-7$; $p_{ap} = 1e-3$; intrinsic QBER 0.005; detector multiplier 1.0	§9.3	NV	CFR B-07; EV-1b (06 baseline column) + EV-1a (03 baseline block)	Assumed constants, not measured values (see CL-45, CL-46).
CL-26	Protocol profile: efficient-BB84, WCP downlink, 1 signal + 2 decoy intensities (one vacuum), finite-key per Sidhu-family fixture	§8, §9.3	NV	CFR B-03; EV-3 + model source inspection	Fixed protocol; no protocol change is claimed anywhere.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-27	Boundary states: physi-cal_characterization NOT-EXECUTED; physi-cal_validation NOT-EXECUTED; hardware_in_loop BLOCKED; laser INHIBITED; tabuk_run NOT-RUN/ NONE; key_release QUARAN-TINED/ ZERO-RELEASED; release PRIVATE-BLOCKED	§1, §16	NV	CFR B-01/B-02; EV-1a (03) + EV-3 (08 AUD-016-020)	Zero physical characterization; zero hardware-in-loop; no experimental run; no released secret key.
CL-28	Margin equation $M = s_{X,0} + s_{X,1} \cdot [1 - h2(_X)] - _EC - 6 \cdot \log2(21/_s) - \log2(2/_c)$ is the Lim et al. 2014 structure in the Sidhu et al. 2022 satellite fixture lineage	§9.3	TS	D3 §2/§6 (Lim PRA 89, 022307; Sidhu npj QI 8, 18 –VERIFIED); CFR N-02 (bit-exact transcription); EV-5 + EV-2	Transcription faithful; the equation' s validity inherits all fixture assumptions (CL-46).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-29	The “21” secrecy budget: $6 \cdot \log_2(21/_s)$ decomposes into 21 one-sided estimation/sampling/ hashing terms with effective deviation parameter $_s/21$	§9.3	TS	D6 §4.2 (Lim 2014 supp. Eqs. (11)–(14), verified); D3 §6	The 21-split validator accepts the fixture’ s internal $= \ln(21/_s)$ form; unscaled $_s$ as effective deviation is rejected.
CL-30	Hash-chain status: 9 of 10 controlled artifacts content-hash-verified (2 direct, 7 after deterministic PDF-repair); artifact 01 (model source) compiles but is not byte-reconstructible	§10.6	NV	CFR §0; audit D1 §A.1; EV-1a/ EV-1c	Artifact 01 carries EV-4/ EV-1c; no claim of byte-level hash verification may be made for it.
CL-31	Controlled-input index: 17/17 controlled-input hashes match the verified manifest; bundle loss file 02 is byte-identical to the V0.6 controlled input FS_loss_XI0.csv	§10.6	NV	CFR §0; EV-3 (10 09)	Independent corroboration of AUD-016-015.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-32	Frozen loss curve: 693 samples, $t = +346 \rightarrow -346$ s strictly decreasing, elevation/efficiency mirror-symmetric about $t = 0$, $t = 0$ present	§10.2	NV	CFR §0 row 2; EV-1a (02 after deterministic repair)	“Monotonic in
CL-33	Profile A (hardened Lim/Sidhu fixture + assumption ledger + statistics upgrade) is the V0.17-TA1 shipping proof profile	§13	AD	D4 §4 recommendation 1; D6 §2/§6	Recommendation, not a security result. All Profile A outputs remain
CL-34	Profile B (Tupkary et al., arXiv:2601.18035, 2026) is the designated V0.18+ upgrade-path anchor, with the unified source-imperfection framework (Optica Quantum 3, 525, 2025) for joint flaw coverage	§13	AD / LS	D4 §4 recommendation 2; CFR L-09; EV-5	CONDITIONAL-COMPUTATION-labeled while characterization is absent. PREPRINT-LABELED: the anchor is a preprint under review; an earlier “published as Quantum 10, 2037” claim was removed as unverified (fail-closed, red-team C1). Its own abstract frames imperfection integration as future work (F-03 amendment).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-35	Target layered architecture: Layer 0 frozen V0.16 fixture → Layer 1 hardened analytic finite-key (Profile A / V0.17) → Layer 2 implementation-security proof extensions (Profile B / V0.18+) → Layer 3 characterization/certification composition	§13	AD	D4 §4 recommendation 5; D6 §11	Layer 3 execution is BLOCKED pending characterization; no layer emits a certified-security claim today.
CL-36	Security-budget composition: $_total = _c + _s + _char (+ _auth)$, with $_char \ \Sigma_j \ _j$ defined once; only joint bounds $\Pr[\text{approve insecure}] \ _char + _protocol$ are valid claim language	§9.4, §14	TS	D6 §4.1–4.3; Tan–Nahar PRX Quantum 7, 020342 (2026) – VERIFIED	Conditional-on-approval language is prohibited. $_char$ and $_auth$ are symbolic (CL-37).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-37	Characterization-budget and symbolic-parameter claims (Δ_{PR} , (,) bounds, $_spf$, $I_{iso}/_out$, $_d$ maps, afterpulse kernel, $_char$, $_auth$, $_varlen$, aging envelopes)	§14, §18	CR	D6 §3, §4.1, §5	All entries SYMBOLIC ONLY – CHARACTERIZATION REQUIRED; no numerical value is assigned to any device parameter; the architecture is a specification, not an authorization to measure.
CL-38	MDI-QKD is architecturally excluded from the Q-Orbit proof path (no relay between two senders in a direct downlink; ground receiver is the party whose trust is at issue)	§13	TS	D4 §2 verdict	Architectural argument; MDI logged as an alternative concept only.
CL-39	Effect 1 (incomplete phase randomization): UNMAPPED-PROOF-REQUIRED as-is; PROOF-PROFILE-CANDIDATE exists (Nahar PR Applied 20, 064031, 2023) gated by phase-PDF characterization	§12	CR	D5 §2 row 1	No phase measurement exists (CFR B-01).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-40	Effect 2 (pulse-to-pulse encoding correlations): UNMAPPED-PROOF-REQUIRED + UNMAPPED-CHARACTERIZATION-REQUIRED	§12	CR	D5 §2 row 2	Breaks IID/Serfling statistics; martingale route (Azuma/Kato) identified; no correlation-length data exist.
CL-41	Effect 3 (intensity correlations): PROOF-PROFILE-CANDIDATE (Zapatero 2021; Sixto 2022; Pereira QST 2025) gated by UNMAPPED-CHARACTERIZATION-REQUIRED	§12	CR	D5 §2 row 3	TH-PAR-004/005 touch only the adjacent independent-fluctuation effect (PARTIAL-SCALAR-STRESS-ONLY).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-42	Effect 4 (state-preparation flaws, basis-dependent part): UNMAPPED-PROOF-REQUIRED; PROOF-PROFILE-CANDIDATE exists (loss-tolerant / unified framework); basis-independent component folded into TH-PAR-008 is PARTIAL-SCALAR-STRESS-ONLY	§12	CR	D5 §2 row 4	GLLP Δ bolt-on covers only basis-independent flaws; rejected as sufficient coverage (D4 §5).
CL-43	Effect 5 (source leakage / Trojan horse): BLOCKING while isolation unmeasured; passive component UNMAPPED-SECURITY-BUDGET + UNMAPPED-CHARACTERIZATION-REQUIRED	§12	BL (active) / CR (passive)	D5 §2 row 5	Strictest resolution governs. No security claim may be emitted while leakage/isolation fields are empty (D5 §4 watchlist #8).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-44	Effect 6 (dead time / recovery): UNMAPPED-SECURITY-BUDGET (strictest of UNMAPPED-MODEL-REQUIRED + USB); TH-PAR-002 touch PARTIAL-SCALAR-STRESS-ONLY	§12	CR	D5 §2 row 6	Dead-time attack literature verified; adversarial treatment partial (preprint-level).
CL-45	Effect 7 (saturation): BLOCKING above the linear regime (detector-control class entry); UNMAPPED-MODEL-REQUIRED (engineering)	§12	BL	D5 §2 row 7	Fixture count model is linear in ; rollover structurally inexpressible.
CL-46	Effect 8 (detector timing jitter): UNMAPPED-CHARACTERIZATION-REQUIRED (proof machinery exists; inputs unmeasured); TH-PAR-002 touch PARTIAL-SCALAR-STRESS-ONLY	§12	CR	D5 §2 row 8	Time-shift enabler if inter-detector/basis asymmetric.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-47	Effect 9 (history-dependent afterpulsing): UNMAPPED-PROOF-REQUIRED; TH-PAR-007 scalar is equilibrium-IID-marginal only (PARTIAL-SCALAR-STRESS-ONLY)	§12	CR	D5 §2 row 9	Detector-side correlated-afterpulse finite-key proof is a genuine open literature problem (D5 §6.1); worst-case-IID pessimism theorem must not be assumed.
CL-48	Effect 10 (detection-efficiency mismatch): PROOF-PROFILE-CANDIDATE gated by UNMAPPED-CHARACTERIZATION-REQUIRED; TH-PAR-002 touch PARTIAL-SCALAR-STRESS-ONLY	§12	CR	D5 §2 row 10	Canonical detector-side proof gap; bounded-mismatch proofs need measured $_min/ _max$.
CL-49	Effect 11 (wavelength-dependent response): UNMAPPED-SECURITY-BUDGET (out-of-band response is an unmodeled Eve→receiver channel)	§12	CR	D5 §2 row 11	Remedy is certified spectral filtering as a device assumption —a security-budget item, not a proof-profile swap.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-50	Effect 12 (polarization-dependent response): PROOF-PROFILE-CANDIDATE + UNMAPPED-CHARACTERIZATION-REQUIRED	§12	CR	D5 §2 row 12	Common-mode-only is a measured property, not an assumption.
CL-51	Effect 13 (detector memory, general cross-pulse effects): UNMAPPED-PROOF-REQUIRED	§12	CR	D5 §2 row 13	Complete correlated-memory detector treatment for this fixture is absent from verified literature (D5 §6.4).
CL-52	Effect 14 (characterization uncertainty): UNMAPPED-CHARACTERIZATION-REQUIRED (definitional)	§12	CR	D5 §2 row 14	At zero characterization every fixture output (incl. CFR N-02) is a conditional computation supporting no security claim.
CL-53	Effect 15 (aging / cross-instance drift): UNMAPPED-CHARACTERIZATION-REQUIRED	§12	CR	D5 §2 row 15	Parameters UNCHARACTERIZED beyond the last characterized epoch absent a characterized aging model.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-54	Master-matrix verdict: 0 of 15 effects MAPPED-IN-CURRENT-FIXTURE; 5 rows UNMAPPED-PROOF-REQUIRED (1, 2, 4, 9, 13); 2 rows BLOCKING (5, 7); 2 rows UNMAPPED-SECURITY-BUDGET (6, 11); remaining 6 rows PROOF-PROFILE-CANDIDATE gated (3, 10, 12) or UNMAPPED-CHARACTERIZATION-REQUIRED (8, 14, 15)	§12	CR	D5 §7	Red-team C5 correction already applied in D5 (row-4 count and row-11 USB label). Fixture' s own 16-row/ 7-unmapped register (AUD-016-010) row-level verification is REQ-05-gated.
CL-55	Any operational reading of the screen outputs (e.g. “33.79% of conditions yield key” as availability/ yield/reliability)	§11.3, §15	OU	CFR N-07/N-08 disclosures; Closure Record §6.3	The fraction is an upper-envelope deterministic screen quantity under per-point window re-optimization; never a probability, reliability, availability, yield, or mission-success figure.

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-56	Released secret key / key availability	§1, §16, §20	BL	CFR B-01/B-02; key_release_status QUARANTINED/ZERO-RELEASED (EV-1a + EV-3)	Prohibited-claims register; EV-10. No key exists to release.
CL-57	Physical validation / experimental confirmation of any model output	—	BL	CFR B-01/B-02; Closure Record §6.6; EV-10	PVR-class activity not performed; gated by characterization campaigns that do not exist.
CL-58	Implementation security / certified device security	—	BL	CFR §4; D5 §7; EV-10	Zero characterization every security-relevant output is conditional; effects 5 and 7 are BLOCKING.
CL-59	Mission success probability	—	BL	CFR §4; Closure Record §6.6; EV-10	Prohibited-claims register.
CL-60	QKD availability (site, link, or service)	—	BL	CFR §4; EV-10	Prohibited-claims register; see CL-55.
CL-61	Tabuk performance (tabuk_run_status NOT-RUN/NONE)	—	BL	CFR B-02, §4; EV-1a + EV-10	No run exists or is planned in scope.
CL-62	Hardware readiness	—	BL	CFR §4; EV-10	Prohibited-claims register.
CL-63	Procurement tolerance	—	BL	CFR §4; EV-10	Prohibited-claims register; frontier crossings are explicitly not hardware acceptance thresholds (CL-17).

Table 2 – continued

Claim ID	Claim (concise)	Manuscript section	Label	Evidence	Notes / disclosures
CL-64	Deployability	—	BL	CFR §4; EV-10	Prohibited-claims register.
CL-65	Field readiness	—	BL	CFR §4; EV-10	Prohibited-claims register.

Ledger row count: 65 claims (CL-01...CL-65).

3. Prohibited-claims register (CFR §4, restated, binding)

The following claim classes are prohibited in every Q-Orbit artifact and downstream document: **mission success probability; QKD availability; Tabuk performance; implementation security; certified device security; procurement tolerance; hardware readiness; deployability; field readiness; released secret key.**

Item	Statement
Violation check	Package-wide sweep result: ZERO violations. Two independent hostile red-team rounds (Round 1: C1–C16; Round 2, Agent G five-persona final review: F-01...F-09) —cumulative 25/25 findings dispositioned FIXED with verification; Round 2’ s prohibited-claims sweep found zero violations; 0 BLOCKING OPEN ISSUE remains. Source: Phase 1 Closure Record §5; Final Red-Team Report §1/§7.

4. Open required inputs (REQ register; all OPEN at 2026-08-27)

Blocker ID	Missing artifact (hash-recorded)	Status	What it gates
REQ-01	controlled_inputs/v0.6/ Q-Orbit_CASE- S1_Run_Manifest_V0.6.json (0714d6e7...)	OPEN	End-to-end re-execution of the supplied model; individual _s/_c, intensities, probabilities, channel-config values (gates CL-09, CL-10, CL-21 full closure).
REQ-02	controlled_inputs/v0.6/ CASE-S1-REF- 001_Baseline_Run_V0.6.json (3673acf4...)	OPEN	Direct confirmation of the V0.6 expected-baseline fixture used by REG-001 ...007 (CL-18).

Table 4 – continued

Blocker ID	Missing artifact (hash-recorded)	Status	What it gates
REQ-03	controlled_inputs/v0.7/ Q-Orbit_CASE- M1_Parameter_and_Provenance_Register_V0.7.csv (fb07b900...)	OPEN	Verification that screen ranges (1e-7...2e-6; 0.003...0.015) are exactly the V0.7 register values; provenance of the 1–221 s sweep bound (CL-21, CL-23).
REQ-04	Original (non-PDF) bytes of Q- Orbit_Kimi_Core_Research_Input.zip	OPEN	Byte-level hash closure on artifacts 1, 2, 4–8, 10 (upgrades CL-30’ s residual EV-1c item).
REQ-05	Five data_processed/ registers (28b9fef2..., 9907aa34..., bc864414..., a008f748..., fb9260f3...)	OPEN	Row-level verification of grid boundary, the 16-row/7-unmapped proof mapping (CL-54), claim/gate registers.

No REQ item is closed by inference. The former computational-input blocker (“V0.16 computational package absent”) is the only CLOSED entry — superseded by the received and hash-verified 10-artifact bundle (Closure Record §3).

End of QO-LEDGER-001.